John Tonn

Interests: Computer Architecture, Digital Design, GPU Architecture, Memory Systems
Looking for an internship summer 2027 (Currently applying to graduate school starting in Fall 2027)

Education

Iowa State University — Ames, IA
Double Major: Computer Engineering, Mathematics — 08/2022 - Present
GPA: 3.84 — Anticipated graduation: May 2027

Professional Experience

Iowa State University — Ames, IA
REU Researcher — 05/2026 - 09/2026

- Characterized the SKY130 library for voltages below 1.8V
- Analyzed efficiency per joule of designs at lower voltages with characterized library
- Learned Cadence Liberate, Cadence Virtuoso, and Cadence Pegasus CLI to generate a repeatable flow
- Post-processed on GDS and netlist files to strengthen confidence in results
- Presented findings at Iowa State Undergraduate Research Showcase

Iowa State University — Ames, IA
Embedded Systems TA — 08/2024 - Present

- Help students familiarize with the C language in an embedded environment on Tiva C123GH6PM microcontroller
- Guided 200+ students in debugging memory management and stack/heap behavior in C
- Explained concepts like microarchitecture decisions and MMIO

Garmin — Olathe, KS
Software Engineering Intern — 05/2025 - 08/2025

- Developed driver-level firmware in C for IMU sensor communication over I2C for the Spy Pole product
- Implemented interrupt-driven data acquisition and state-machine-based device control
- Debugged hardware/firmware interaction using logic analyzers and on-target debugging

Projects

Senior Design — 08/2025 - Present
Embedded GPU for Fabrication

- Designed a predicated programmable pipeline GPU in Verilog supporting vertex and fragment shading on a custom ISA
- Developed SystemVerilog testbench and coverage-driven verification strategy
- Synthesized design using OpenLane; achieved 40 MHz at 10mm^2 area
- Evaluated performance bottlenecks in memory bandwidth and pipeline hazards

Activities and Leadership

Chip Forge — 08/2024-Present
President — 05/2026-Present

- Refined ASIC design and verification procedures for team members
- Mentored students in RTL design, synthesis, and ASIC flow using OpenLane
- Project lead for the entire ASIC fabrication process
- Let post-silicon validation for returned designs

Cyclone Rocketry — 08/2022-Present
Avionics Team Lead — 08/2024-08/2025

- Designed and flew embedded rocket computers to predict and control the flight of a rocket
- Implemented a robust development cycle to find faults and promote project growth
- Designed flight control systems achieved getting a rocket to 9,966 ft with a goal of 10,000

Technical Skills

- Software: C, C++, Python, Java, C#, MIPS, RISC-V
- HDL: Verilog, VHDL, SystemVerilog
- Tools: Openlane, ModelSim, Vivado, GTKWave